

Aharonov–Bohm Signature of a Field Response

Sina Montazeri¹

¹ *University of North Texas*

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A family of rank-three configurations on a flat three-torus exhibits global holonomy dependence in the curvature Hessian of a field. Every member has vanishing assembled curvature, and all members are gauge equivalent on every contractible region, while their global gauge holonomies may differ. A preserved internal line identifies a rank-one gravitational summand of the field response. The gauge-reduced curvature Hessian has an infinite-dimensional kernel and a non-trace-class heat semigroup, while the rank-one tangential factor determines a translation-invariant reducing subspace on which $P_\eta L_{\theta,k} e^{-tL_{\theta,k}}$ is trace class. Its exact Fourier expansion contains the product of the gauge holonomy character and the holonomy character of the preserved rank-one assembled connection. At $k = 1/6$ and $t = 1$, the configurations with $\theta = 0$ and $\theta = 1/2$ have weighted heat traces separated by more than 0.9731179060, and the trace varies nontrivially along the rank-one gravitational summand. Thus locally identical curvature data need not determine the curvature Hessian; the resulting distinction has the local–global form of the Aharonov–Bohm problem.

LOCAL CURVATURE AND GLOBAL HOLONOMY

The Aharonov–Bohm effect established that vanishing local field strength does not determine the physical content of a gauge connection [1]. In its familiar formulation, the phase of a quantum probe detects holonomy around an inaccessible region. The present model has the same local-curvature/global-holonomy distinction in the curvature Hessian determined by a field. The spectral theory of Aharonov–Bohm operators studies how eigenvalues of magnetic Schrödinger operators depend on flux and pole position [2, 3]; here the holonomy dependence appears instead in the Hessian of a curvature energy, and the trace carries the product of a gauge character and a gravitational one.

Such a distinction occurs for a field whose configurations consist of a metric internal connection A and a symmetric internal-valued tensor K [4]. These two variables form a single metric connection on the direct sum of the tangent and internal bundles. Earlier geometric constructions placed gravitational and gauge variables within a higher-dimensional metric or a de Sitter connection [5, 6]. The squared curvature of the assembled connection determines both the response of a configuration and the curvature Hessian at a critical configuration. On a flat three-torus, the family (A_θ, K_k) has members that are gauge equivalent on every contractible region but carry distinct global holonomy. The connection A_θ preserves the internal image line of K_k , so variation of k belongs to the preserved rank-one gravitational summand.

The self-adjoint representative of the gauge-reduced curvature Hessian has an infinite-dimensional kernel, and its ordinary heat trace therefore diverges. The rank-one tangential factor of K_k determines a compact translation group and a closed infinite-dimensional reducing subspace. On that subspace the bounded function

$$h_t(L) = L e^{-tL}, \quad t > 0,$$

defines a trace-class operator whose exact trace has Fourier coefficients proportional to

$$\cos(2\pi m\theta) \cos(2\pi mk).$$

The first factor is the character of the transverse gauge holonomy and the second is the character of the holonomy of the preserved rank-one assembled connection. Their product arises in the spectrum of the curvature Hessian. A bound on the complete Fourier coefficient, including the term independent of k , proves that the flat configurations $(A_0, K_{1/6})$ and $(A_{1/2}, K_{1/6})$ have different weighted heat traces at $t = 1$.

The field determines the configurations (A_θ, K_k) , their curvature energy and response, and the curvature Hessian at each configuration. The weighted heat trace shows that this Hessian can retain global holonomy information absent from the local curvature. The next two sections define the field and the flat family, after which the complete spectrum on the reducing subspace gives its Fourier expansion. The appendices supply the corresponding finite-matrix and Poisson-summation calculations.

THE FIELD AND ITS CURVATURE HESSIAN

Let M carry a Riemannian metric h , and let the finite-rank oriented bundle $N \rightarrow M$ carry a metric g_N . A configuration is

$$\Psi = (\nabla^N, K), \quad K \in \Gamma(\text{Sym}^2 T^*M \otimes N),$$

where ∇^N is metric. Contraction with $X \in TM$ gives $K_X Y = K(X, Y)$, and the two metrics determine the adjoint $K_X^\dagger : N \rightarrow TM$. The base Levi–Civita connection and the configuration form the metric connection

$$\nabla_{\Psi, X}^\oplus = \begin{pmatrix} \nabla_X^h & -K_X^\dagger \\ K_X & \nabla_X^N \end{pmatrix} \quad (1)$$

This connection acts on $TM \oplus N$. Preservation of $h \oplus g_N$ forces the signs of the mixed blocks.

Write $\mathcal{F}_\Psi^\oplus$ for the curvature of (1). With the Hilbert–Schmidt metric on its skew endomorphism values, the curvature energy is

$$\Phi(\Psi) = \frac{1}{2} \int_M |\mathcal{F}_\Psi^\oplus|^2 dV_h. \quad (2)$$

The configuration-space metric G_Ψ determines the unique metric-dual response $\mathcal{Y}_\Psi$ by

$$D\Phi_\Psi(v) = G_\Psi(\mathcal{Y}_\Psi, v).$$

The orthogonal splitting into connection and K variations gives $\mathcal{Y}_\Psi = (\mathcal{Y}_{\nabla, \Psi}, \mathcal{Y}_{K, \Psi})$, with the connection component giving the gauge response. Whenever

$$L_K = \text{im } K \subset N$$

is an oriented metric line that ∇^N preserves, the assembled connection splits as

$$TM \oplus N = (TM \oplus L_K) \oplus L_K^\perp.$$

Variations of K within L_K then reproduce the rank-one Gauss–Codazzi response of the first summand. The gravitational interpretation applies to this projected rank-one component.

At the critical configurations considered below, the second variation of (2) defines a densely defined, closed nonnegative quadratic form on configuration variations modulo internal gauge directions. When $\mathcal{F}_\Psi^\oplus = 0$, this Hessian form has the particularly direct expression

$$Q_\Psi([v]) = \|\mathcal{L}_\Psi v\|_{L^2}^2, \quad \mathcal{L}_\Psi v = D_{\nabla^\oplus} X_v, \quad (3)$$

where the variation v has assembled one-form X_v . The first representation theorem [7, Chap. VI, Sec. 2, Theorem 2.1] assigns to Q_Ψ its unique nonnegative self-adjoint representative L_Ψ . Thus the same curvature energy determines the classical response and the spectral calculus of L_Ψ .

A metric gauge transformation u acts by

$$A^u = uAu^{-1} - duu^{-1}, \quad K^u = uK,$$

in a chosen trivialization of N . It transports $\nabla_\Psi^\oplus$, $\mathcal{L}_\Psi$, and L_Ψ by conjugation, so the spectral traces below are invariant under the induced gauge conjugacies.

FLAT CONFIGURATIONS WITH DISTINCT GLOBAL HOLONOMY

Take

$$M = \mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3, \quad N = \mathbb{T}^3 \times \mathbb{R}^3$$

with their standard metrics and oriented orthonormal frames (t_1, t_2, t_3) and (e_1, e_2, e_3) . For orthonormal vectors u, v , let $J_{uv}u = v$, $J_{uv}v = -u$, and let J_{uv} vanish on the orthogonal complement. Put $J = J_{e_2 e_3}$ and consider

$$A_\theta = \theta dx^1 J, \quad K_k = k dx^1 \otimes dx^1 \otimes e_1, \quad k \neq 0. \quad (4)$$

The connection one-form and the tangential factor of K_k both lie along dx^1 . The spectral calculation below treats this aligned family.

Every block of the assembled curvature vanishes. Since A_θ is constant and takes values in one generator,

$$F_{A_\theta} = dA_\theta + \frac{1}{2}[A_\theta \wedge A_\theta] = 0$$

and $Je_1 = 0$ gives $D_{A_\theta} K_k = 0$. Every nonzero map $(K_k)_X : TM \rightarrow N$ is a multiple of the same rank-one map, which gives

$$-K_X^\dagger K_Y + K_Y^\dagger K_X = 0, \quad -K_X K_Y^\dagger + K_Y K_X^\dagger = 0.$$

These are all blocks of the assembled curvature, hence

$$\mathcal{F}^\oplus(A_\theta, K_k) = 0, \quad \Phi(A_\theta, K_k) = 0. \quad (5)$$

Every configuration in (4) is therefore a critical global minimum of the nonnegative curvature energy.

On a contractible region $U \subset \mathbb{T}^3$, the gauge transformation

$$u_{\theta, U}(x) = \exp(\theta x^1 J)$$

is single valued, fixes e_1 , and sends (A_θ, K_k) to $(0, K_k)$. It is globally single valued precisely when $\theta \in \mathbb{Z}$. The first-circle gauge holonomy is

$$H_g(\theta) = \exp(2\pi\theta J)|_{E_K}, \quad E_K = \text{span}\{e_2, e_3\}, \quad (6)$$

so $\theta \in \mathbb{R}/\mathbb{Z}$, with conjugacy character

$$\frac{1}{2} \text{Tr}_{E_K} H_g(\theta)^m = \cos(2\pi m\theta).$$

The connection A_θ preserves the image line $L_K = \mathbb{R}e_1$. Consequently $TM \oplus L_K$ is a rank-one assembled subsystem and variation of k lies in the gravitational summand defined above. On the preserved plane

$$Q_K = \text{span}\{t_1, e_1\},$$

the assembled connection around the first circle is $kJ_{t_1 e_1} dx^1$. Its holonomy

$$H_K(k) = \exp(2\pi k J_{t_1 e_1})|_{Q_K} \quad (7)$$

has character

$$\frac{1}{2} \text{Tr}_{Q_K} H_K(k)^m = \cos(2\pi mk).$$

Equations (6) and (7) are conjugacy invariant and arise from different commuting blocks of the same flat assembled connection.

Finally, the rank-one tangential factor of K_k determines $\eta = dx^1$ up to sign and the compact group $H_\eta \cong \mathbb{T}^2$ of translations in x^2, x^3 . Haar averaging defines

$$P_\eta v = \int_{H_\eta} T_y v dy. \quad (8)$$

The constant coefficients in (4) imply that P_η commutes with the linearized curvature map, the gauge action, and $L_{\theta,k}$. The tensor K_k therefore determines, before the spectral calculation, the closed infinite-dimensional reducing subspace of all modes e^{inx^1} .

SPECTRUM AND WEIGHTED HEAT TRACE OF THE CURVATURE HESSIAN

Complexify E_K and choose J -eigenvectors q_σ satisfying $Jq_\sigma = \sigma i q_\sigma$, where $\sigma = \pm 1$. Flux-dependent spectra on a ring provide the corresponding physical precedent [8, 9]. On the mode $e^{inx^1} q_\sigma$, the connection in Eq. (4) gives the covariant momentum

$$p_{n,\sigma} = n + \sigma\theta. \quad (9)$$

The variation in each σ -eigenspace contains three connection coefficients and six symmetric E_K -valued shape coefficients. Appendix A reduces their quadratic form to elementary 2×2 blocks and obtains the characteristic polynomial of the mode Gram matrix $G(p, k)$,

$$\begin{aligned} \det(\lambda - G(p, k)) = & -\lambda^2(2p^2 - \lambda)^3 \\ & \times ((p-k)^2 - \lambda)^2((p+k)^2 - \lambda)^2. \end{aligned} \quad (10)$$

The two zero directions consist of one tangent to the internal-gauge orbit and one non-gauge direction. After the gauge quotient, the nonzero spectrum and multiplicities remain

$$\begin{array}{c|ccc} \lambda & 2p^2 & (p-k)^2 & (p+k)^2 \\ m(\lambda) & 3 & 2 & 2. \end{array} \quad (11)$$

The spectrum also contains the additional zeros at $p = 0$ or $p = \pm k$.

Appendix A gives one nonzero kernel class in every Fourier subspace of the gauge-reduced curvature Hessian. Thus $\dim \ker L_{\theta,k} = \infty$, its resolvent is noncompact, and

$$\text{Tr}(e^{-tL_{\theta,k}}) = \infty.$$

The bounded function $h_t(x) = xe^{-tx}$ vanishes on the entire kernel. Equations (9) and (11) therefore give the

weighted heat trace

$$\begin{aligned} \Xi_{\theta,k}(t) &= \text{Tr}(P_\eta L_{\theta,k} e^{-tL_{\theta,k}}) \\ &= \sum_{\sigma=\pm 1} \sum_{n \in \mathbb{Z}} \left[6p_{n,\sigma}^2 e^{-2tp_{n,\sigma}^2} \right. \\ &\quad \left. + 2(p_{n,\sigma} - k)^2 e^{-t(p_{n,\sigma} - k)^2} \right. \\ &\quad \left. + 2(p_{n,\sigma} + k)^2 e^{-t(p_{n,\sigma} + k)^2} \right]. \end{aligned} \quad (12)$$

On compact parameter sets, the bound $C(1+n^2)e^{-cn^2}$ controls the summand. Hence $P_\eta h_t(L_{\theta,k})$ is trace class, and dominated convergence permits termwise differentiation.

For $c > 0$, define

$$\mathcal{A}_c(m, t) = \sqrt{\frac{\pi}{c}} t^{-3/2} \left(\frac{1}{2} - \frac{\pi^2 m^2}{ct} \right) e^{-\pi^2 m^2/(ct)}. \quad (13)$$

Appendix B derives from Poisson summation that

$$\begin{aligned} \Xi_{\theta,k}(t) &= \sum_{m \in \mathbb{Z}} \left[6\mathcal{A}_2(m, t) \cos(2\pi m\theta) \right. \\ &\quad \left. + 8\mathcal{A}_1(m, t) \cos(2\pi m\theta) \cos(2\pi mk) \right]. \end{aligned} \quad (14)$$

The zero-winding term is

$$\Xi^{(0)}(t) = \frac{\sqrt{\pi}}{2t^{3/2}} (3\sqrt{2} + 8), \quad (15)$$

which is independent of both holonomies. For $m \neq 0$, define the mixed holonomy contribution

$$W_m^{\theta k} = 8\mathcal{A}_1(m, t) \cos(2\pi m\theta) \cos(2\pi mk).$$

It is the product of the characters in (6) and (7). To establish a distinction of the total trace, the complete Fourier coefficient must retain a definite sign. At $t = 1$, Appendix B proves for every $m \neq 0$ that

$$\frac{|\mathcal{A}_1(m, 1)|}{|\mathcal{A}_2(m, 1)|} < \frac{3}{4}. \quad (16)$$

Since both coefficients are negative, this gives, for every k ,

$$B_m(k) := 6\mathcal{A}_2(m, 1) + 8\mathcal{A}_1(m, 1) \cos(2\pi mk) < 0. \quad (17)$$

Consequently

$$\Xi_{0,k}(1) - \Xi_{1/2,k}(1) = 2 \sum_{m \in 2\mathbb{Z}+1} B_m(k) < 0. \quad (18)$$

Thus $(A_0, K_{1/6})$ and $(A_{1/2}, K_{1/6})$ are curvature-flat and gauge equivalent on every contractible region, while their weighted heat traces differ. This is an Aharonov–Bohm-type distinction in the curvature Hessian.

Appendix B derives rigorous enclosures from finite lattice truncations and explicit Gaussian tail bounds,

$$\begin{aligned}\Xi_{0,1/6}(1) &= 10.3631980884\dots, \\ \Xi_{1/2,1/6}(1) &= 11.3363159945\dots,\end{aligned}\quad (19)$$

with

$$\Xi_{1/2,1/6}(1) - \Xi_{0,1/6}(1) > 0.9731179060. \quad (20)$$

Differentiating the absolutely convergent series along the preserved rank-one gravitational summand gives

$$\partial_k \Xi_{\theta,k}(t) = -16\pi \sum_{m \in \mathbb{Z}} m \mathcal{A}_1(m, t) \cos(2\pi m \theta) \sin(2\pi m k). \quad (21)$$

For $k = 1/6$ and $t = 1$, its first Fourier coefficient in θ is nonzero. Hence $\partial_k \Xi_{\theta,1/6}(1)$ is nonzero on a nonempty open subset of the holonomy circle. At $\theta = 1/6$, the interval in Appendix B gives

$$\partial_k \Xi_{1/6,1/6}(1) = 0.0373922800861\dots > 0. \quad (22)$$

Whenever $\mathcal{A}_1(m, t) \neq 0$,

$$\frac{W_m^{\theta k}}{8\mathcal{A}_1(m, t)} = \cos(2\pi m \theta) \cos(2\pi m k), \quad (23)$$

$$\frac{\partial_k W_m^{\theta k}}{-16\pi m \mathcal{A}_1(m, t)} = \cos(2\pi m \theta) \sin(2\pi m k). \quad (24)$$

At $m = 1$ and $\theta = k = 1/6$, these two dimensionless quantities are

$$\frac{1}{4}, \quad \frac{\sqrt{3}}{4}. \quad (25)$$

THE AHARONOV–BOHM DISTINCTION

The configurations $(A_0, K_{1/6})$ and $(A_{1/2}, K_{1/6})$ have identical local curvature and are gauge equivalent on every contractible region, while (18) proves that their weighted heat traces differ. Because $L_{\theta,k}$ represents the Hessian of the single assembled curvature energy, its Fourier coefficient contains both the gauge holonomy character $\cos(2\pi m \theta)$ and the character $\cos(2\pi m k)$ of the preserved rank-one assembled connection. Equation (21) shows that the trace changes along the gravitational k -variation even though the background assembled curvature vanishes.

This result places the local–global content of the Aharonov–Bohm problem in the curvature Hessian of the field. Heat traces classically encode geometric information [10, 11], and for flat bundles they involve the holonomy representation [12]. The weighted trace here is determined by the curvature Hessian of a single assembled connection, and its Fourier coefficients carry the product of the gauge holonomy character and the holonomy character of the preserved rank-one gravitational summand.

The gauge-invariant weighted heat trace is intrinsic to the model, while particle trajectories, quantum states, interference experiments, interaction terms, and measurement apparatus would belong to a physical realization of the distinction.

The ordinary heat semigroup is not trace class because the gauge-reduced Hessian has an infinite-dimensional kernel. The function $h_t(x) = x e^{-tx}$ vanishes at zero, so $P_\eta h_t(L_{\theta,k})$ is trace class while the kernel remains part of the Hilbert space. The parameter $t > 0$ indexes this family of spectral functions, and the model determines the complete map $t \mapsto \Xi_{\theta,k}(t)$.

Equations (23)–(25) are exact dimensionless relations of the torus model. Empirical comparison would require a physical realization, preparation of θ and k , identification of the Hessian modes with measured quantities, calibration of t , and a measurement map. Mass gaps, vacuum energies, and temporal evolution require further structures beyond this model.

The field therefore contains its configurations and local curvature together with the law governing admissible variations, whose Hessian can retain global holonomy information. In this model the distinction is quantitative. Gauge holonomy changes a weighted heat trace of the curvature Hessian, and that trace varies within the preserved rank-one gravitational summand.

Mode spectrum of the gauge-reduced curvature Hessian

The combined connection–shape variation space gives the complete spectrum. Complexify E_K and fix a J -eigenvalue σi . With

$$B = J_{t_1 e_1}, \quad M_\sigma = J_{e_1 q_\sigma}, \quad S_{\ell, \sigma} = J_{t_\ell q_\sigma},$$

only the following commutators contribute,

$$[B, M_\sigma] = -S_{1, \sigma}, \quad [B, S_{1, \sigma}] = M_\sigma.$$

Write a variation in the σ -eigenspace as

$$\begin{aligned}a &= \sum_{i=1}^3 a_i dx^i M_\sigma, \\ \kappa &= \sum_{i,j=1}^3 \kappa_{ij} dx^i \odot dx^j q_\sigma, \quad \kappa_{ij} = \kappa_{ji}.\end{aligned} \quad (26)$$

Its assembled one-form coefficient is

$$X_i = a_i M_\sigma + \sum_{\ell=1}^3 \kappa_{i\ell} S_{\ell, \sigma}.$$

For a general Fourier mode $e^{in \cdot x}$, put

$$p_i = n_i + \sigma \theta_i, \quad d_i = i p_i.$$

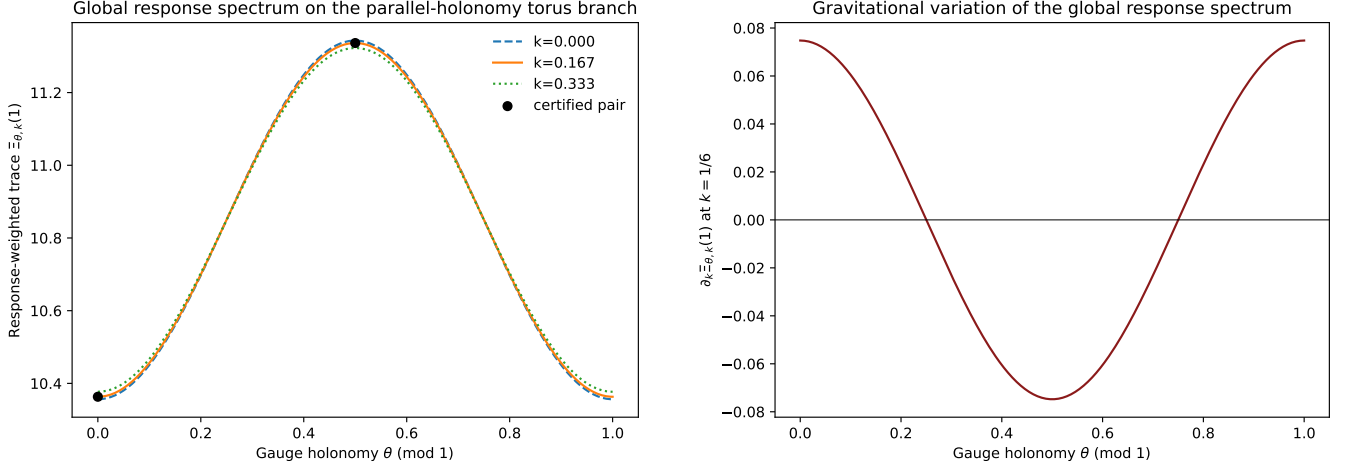


FIG. 1. The weighted heat trace at $t = 1$ for the aligned torus family. Left, $\Xi_{\theta,k}(1)$ as a function of gauge holonomy; black points mark the two rigorously bounded values at $k = 1/6$. Right, $\partial_k \Xi_{\theta,k}(1)$ at $k = 1/6$, corresponding to variation within the preserved rank-one gravitational summand. The curves show the Fourier series; Eqs. (19)–(20) and (22) provide the numerical bounds.

For this kernel calculation, allow a general flat holonomy $(\theta_1, \theta_2, \theta_3)$. The linearized assembled curvature then has, for $i < j$, the twelve coefficients

$$m_{ij} = d_i a_j - d_j a_i + k(\delta_{i1} \kappa_{j1} - \delta_{j1} \kappa_{i1}), \quad (27)$$

$$s_{ij}^1 = d_i \kappa_{j1} - d_j \kappa_{i1} - k(\delta_{i1} a_j - \delta_{j1} a_i), \quad (28)$$

$$s_{ij}^\ell = d_i \kappa_{j\ell} - d_j \kappa_{i\ell}, \quad \ell = 2, 3. \quad (29)$$

Use the following order for the nine independent inputs,

$$(a_1, a_2, a_3, \kappa_{11}, \kappa_{12}, \kappa_{13}, \kappa_{22}, \kappa_{23}, \kappa_{33}).$$

The configuration-space norm and curvature Hessian are

$$\|v\|_G^2 = 2 \sum_i |a_i|^2 + \sum_i |\kappa_{ii}|^2 + 2 \sum_{i < j} |\kappa_{ij}|^2, \quad (30)$$

$$H(v, v) = 2 \sum_{i < j} \left(|m_{ij}|^2 + \sum_{\ell=1}^3 |s_{ij}^\ell|^2 \right). \quad (31)$$

These formulas display the 12×9 mode map and all metric weights.

An infinitesimal internal gauge parameter ξM_σ gives

$$a_i = -d_i \xi, \quad \kappa_{11} = k\xi, \quad \kappa_{ij} = 0 \quad ((i, j) \neq (1, 1)). \quad (32)$$

Substitution into (27)–(29) gives zero. The shape entry in (32) is essential; the connection component alone is not tangent to the gauge orbit when $k \neq 0$.

The kernel also contains a vector transverse to the gauge orbit. In the input order above it is

$$\begin{aligned} w(d, k) = & (-d_1(k^2 + d_1^2), -d_1^2 d_2, -d_1^2 d_3, 0, \\ & -kd_1 d_2, -kd_1 d_3, -kd_2^2, \\ & -kd_2 d_3, -kd_3^2). \end{aligned} \quad (33)$$

Every coefficient in (27)–(29) vanishes after direct substitution. Whenever $w(d, k) \neq 0$, it cannot be tangent to the gauge orbit because its κ_{11} entry is zero, whereas a nonzero vector (32) has $\kappa_{11} = k\xi \neq 0$. If (33) vanishes, the same component formulas instead annihilate

$$a_1 = 1, \quad a_2 = a_3 = 0, \quad \kappa = 0$$

which again represents a non-gauge kernel class. Every Fourier subspace therefore contains a nonzero kernel class after the gauge quotient. Distinct Fourier subspaces are orthogonal, which proves

$$\dim \ker L_{\theta,k} = \infty.$$

On the trigonometric-polynomial core, (27)–(29) define a closable map. Indeed, if $v_r \rightarrow 0$ and $\mathcal{L}v_r \rightarrow z$, then for every smooth test field w ,

$$\langle z, w \rangle = \lim_r \langle \mathcal{L}v_r, w \rangle = \lim_r \langle v_r, \mathcal{L}^\dagger w \rangle = 0.$$

Its graph closure therefore defines a closed nonnegative quadratic form on the Hilbert-space quotient by the closed infinitesimal gauge directions and hence a unique nonnegative self-adjoint representative. The heat semigroup acts as the identity on the infinite kernel, so it cannot be trace class.

The main result imposes the aligned conditions below. On the range of P_η ,

$$\begin{aligned} n_2 = n_3 = 0, & & \theta_2 = \theta_3 = 0, \\ d_1 = ip, & & d_2 = d_3 = 0. \end{aligned}$$

The pairs $i, j = (1, 2)$ and $(1, 3)$ give

$$\begin{aligned} (m_{12}, s_{12}^1) &= (ipa_2 + k\kappa_{12}, -ka_2 + ip\kappa_{12}), \\ (m_{13}, s_{13}^1) &= (ipa_3 + k\kappa_{13}, -ka_3 + ip\kappa_{13}), \end{aligned} \quad (34)$$

$$\begin{aligned} s_{12}^2 &= ip\kappa_{22}, & s_{12}^3 &= ip\kappa_{23}, \\ s_{13}^2 &= ip\kappa_{23}, & s_{13}^3 &= ip\kappa_{33}. \end{aligned} \quad (35)$$

All $(2, 3)$ outputs vanish. The variables a_1, κ_{11} do not appear, giving one gauge-orbit zero direction and one non-gauge zero direction.

Removing the equal input and output weights in each (a_2, κ_{12}) or (a_3, κ_{13}) pair reduces (34) to the matrix

$$C(p, k) = \begin{pmatrix} ip & k \\ -k & ip \end{pmatrix}.$$

Its Gram matrix is

$$C(p, k)^\dagger C(p, k) = \begin{pmatrix} p^2 + k^2 & -2ipk \\ 2ipk & p^2 + k^2 \end{pmatrix},$$

whose eigenvalues are $(p - k)^2$ and $(p + k)^2$. There are two identical blocks. Equation (35), together with the tensor weights in (30)–(31), gives $2p^2$ once from κ_{22} , once from κ_{33} , and once from κ_{23} . Hence

$$\begin{aligned} \det(\lambda - G) &= -\lambda^2(2p^2 - \lambda)^3 \\ &\quad \times ((p - k)^2 - \lambda)^2((p + k)^2 - \lambda)^2, \end{aligned}$$

which proves (10) and every multiplicity in (11) by direct factorization.

Fourier expansion and bounds for the weighted heat trace

The nonzero spectrum in (11) gives, for one J -eigenspace and one covariant momentum p ,

$$6p^2 e^{-2tp^2} + 2(p - k)^2 e^{-t(p - k)^2} + 2(p + k)^2 e^{-t(p + k)^2}. \quad (36)$$

Summing (36) over $n \in \mathbb{Z}$ and $\sigma = \pm 1$ gives (12). Every term is nonnegative. Denote the summand by $s_{n,\sigma}(\theta, k, t)$. On every compact parameter set, suitable constants $C, c > 0$ give the uniform bound

$$0 \leq s_{n,\sigma}(\theta, k, t) \leq C(1 + n^2)e^{-cn^2}.$$

This estimate proves absolute and locally uniform convergence. Multiplication by any fixed polynomial in n, θ, k, t^{-1} preserves Gaussian summability, so all derivatives below also converge locally uniformly.

The Gaussian Poisson identity (see, e.g., [13]) gives

$$\begin{aligned} \sum_{n \in \mathbb{Z}} e^{-tc(n+a)^2} &= \sqrt{\frac{\pi}{ct}} \sum_{m \in \mathbb{Z}} \\ &\quad \times e^{-\pi^2 m^2 / (ct)} e^{2\pi i m a}, \end{aligned} \quad c, t > 0. \quad (37)$$

Both sides and their t -derivatives converge absolutely. Applying $-\partial_t$ to (37) yields

$$\sum_{n \in \mathbb{Z}} c(n+a)^2 e^{-tc(n+a)^2} = \sum_{m \in \mathbb{Z}} \mathcal{A}_c(m, t) e^{2\pi i m a}, \quad (38)$$

with $\mathcal{A}_c$ given by (13).

The first term of (36) equals three times the left-hand side of (38) with $c = 2$ and $a = \sigma\theta$. Summation over σ gives

$$6\mathcal{A}_2(m, t) \cos(2\pi m\theta).$$

The remaining two terms equal twice the $c = 1$ identity with shifts

$$a = \sigma\theta - k, \quad a = \sigma\theta + k.$$

Summing both shifts and both J -eigenspaces gives

$$8\mathcal{A}_1(m, t) \cos(2\pi m\theta) \cos(2\pi mk).$$

This proves (14). At $m = 0$,

$$\mathcal{A}_c(0, t) = \frac{1}{2} \sqrt{\frac{\pi}{c}} t^{-3/2},$$

and (15) follows immediately.

The two holonomy characters follow directly from the two-dimensional rotation matrices,

$$\begin{aligned} \frac{1}{2} \text{Tr}_{E_K} H_g(\theta)^m &= \cos(2\pi m\theta), \\ \frac{1}{2} \text{Tr}_{Q_K} H_K(k)^m &= \cos(2\pi mk). \end{aligned} \quad (39)$$

Thus $W_m^{\theta k}$ is the product of these two conjugacy-invariant characters, multiplied by $8\mathcal{A}_1(m, t)$.

The following estimate shows that the complete Fourier coefficient remains negative after adding the mixed holonomy contribution. At $t = 1$, both $\mathcal{A}_1(m, 1)$ and $\mathcal{A}_2(m, 1)$ are negative for $m \neq 0$. Put $x = \pi^2 m^2 > 9$. Equation (13) gives

$$\frac{|\mathcal{A}_1(m, 1)|}{|\mathcal{A}_2(m, 1)|} = 2\sqrt{2} \frac{x - \frac{1}{2}}{x - 1} e^{-x/2}. \quad (40)$$

For $x > 9$,

$$\frac{x - \frac{1}{2}}{x - 1} < \frac{17}{16}, \quad e^{-x/2} < e^{-9/2} < \frac{1}{10}, \quad 2\sqrt{2} < 3.$$

The middle inequality follows already from $e^{9/2} > 1 + 9/2 + (9/2)^2/2 > 10$. Hence

$$\frac{|\mathcal{A}_1(m, 1)|}{|\mathcal{A}_2(m, 1)|} < \frac{51}{160} < \frac{3}{4},$$

which proves (16). The largest possible absolute value of the mixed holonomy contribution is therefore strictly

smaller than $6|\mathcal{A}_2(m, 1)|$, proving (17). Subtracting the winding series at $\theta = 0$ and $\theta = 1/2$ retains precisely the odd winding numbers with a factor of two. Absolute convergence then proves (18).

For the numerical bounds, retain $|n| \leq 8$ in the lattice series. Monotone Gaussian integral bounds control every omitted shifted second moment, and rational enclosures of the exponential give

$$\begin{aligned} 10.363198088434166516 &< \Xi_{0,1/6}(1), \\ \Xi_{0,1/6}(1) &< 10.363198088434166517, \\ 11.336315994496321845 &< \Xi_{1/2,1/6}(1), \\ \Xi_{1/2,1/6}(1) &< 11.336315994496321846. \end{aligned} \quad (41)$$

The difference of the outer endpoints proves (20). A separate Fourier-series enclosure retaining $|m| \leq 2$, with its own Gaussian tail bound, overlaps each interval in (41); the intervals for the two configurations remain disjoint. The lattice and Fourier calculations also give overlapping positive intervals for the derivative, the narrower of which is

$$0.037392280086118222 < \partial_k \Xi_{1/6,1/6}(1), \quad (42)$$

$$\partial_k \Xi_{1/6,1/6}(1) < 0.037392280086118223. \quad (43)$$

These inequalities are rigorous bounds for the exact infinite sums.

Differentiating its k -dependence gives

$$\partial_k W_m^{\theta k} = -16\pi m \mathcal{A}_1(m, t) \cos(2\pi m \theta) \sin(2\pi m k),$$

which proves (21). The coefficient $\mathcal{A}_1(1, 1)$ is nonzero because

$$\mathcal{A}_1(1, 1) = \sqrt{\pi} \left(\frac{1}{2} - \pi^2 \right) e^{-\pi^2}.$$

At $\theta = k = 1/6$,

$$\cos \frac{\pi}{3} \cos \frac{\pi}{3} = \frac{1}{4}, \quad \cos \frac{\pi}{3} \sin \frac{\pi}{3} = \frac{\sqrt{3}}{4},$$

which verifies (25).

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